

Spark UI, Optimizer & Internal concepts (PhysicalPlan, Jobs, Stages, Tasks)

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This document:
https://github.com/Arnaud-Nauwynck/presentations/
/pres-bigdata/12-sparkui-internals-plan-job-stage-task

Outline

What’s visible from Spark-ui

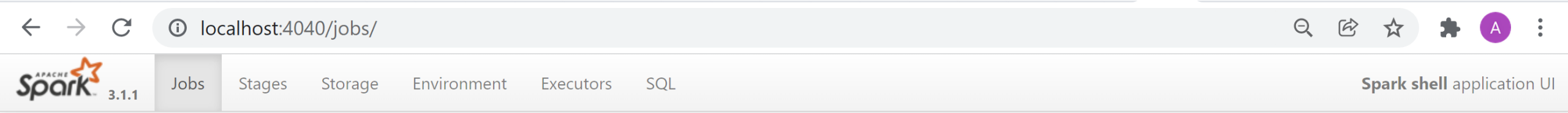
The concepts behinds Dataset/RDD execution : Job, Stage, Task

The architecture of deployment : Client, Driver, Executor

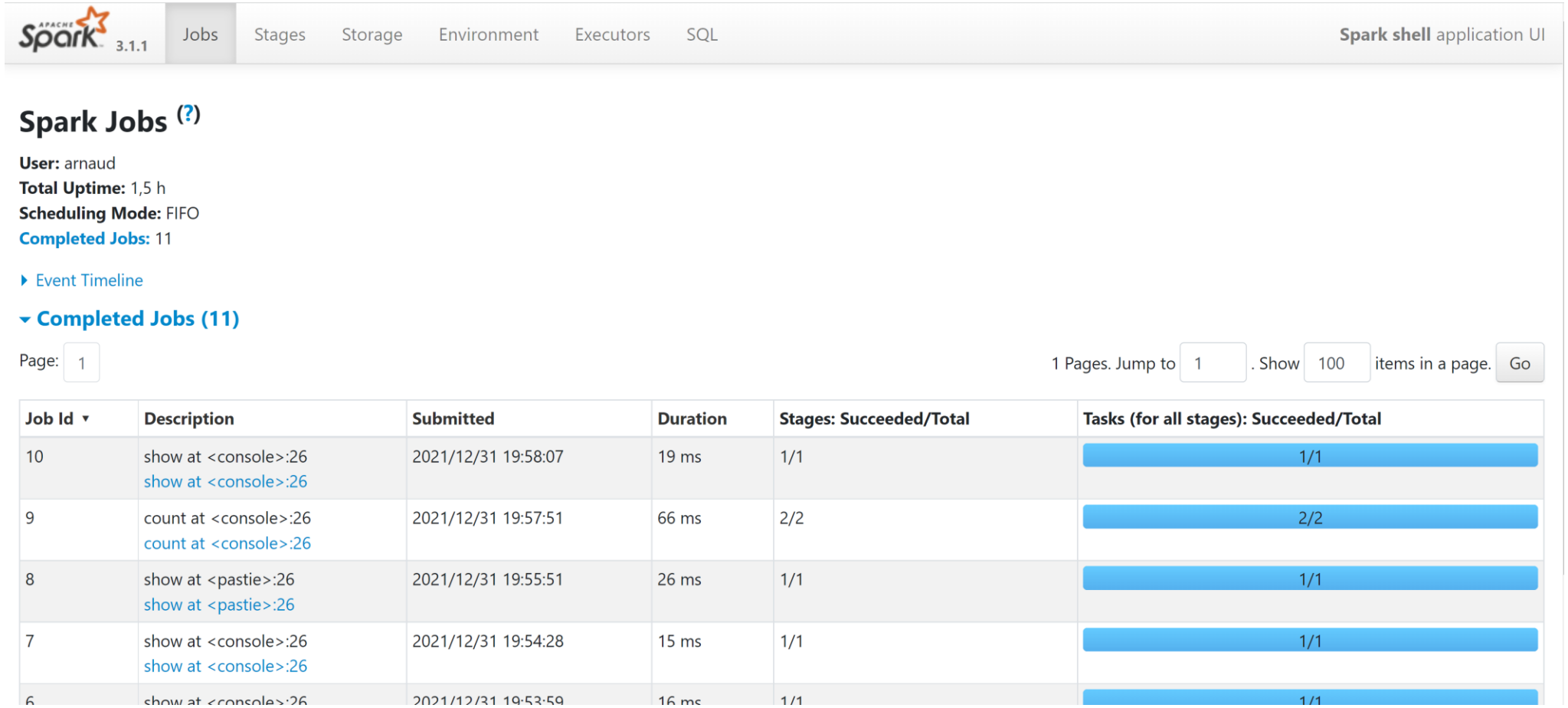
Spark-ui

```
Spark context Web UI available at http://DESKTOP-2EGCC8R:4040
Spark context available as 'sc' (master = local[*], app id = l
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.1.1
      /_/
```

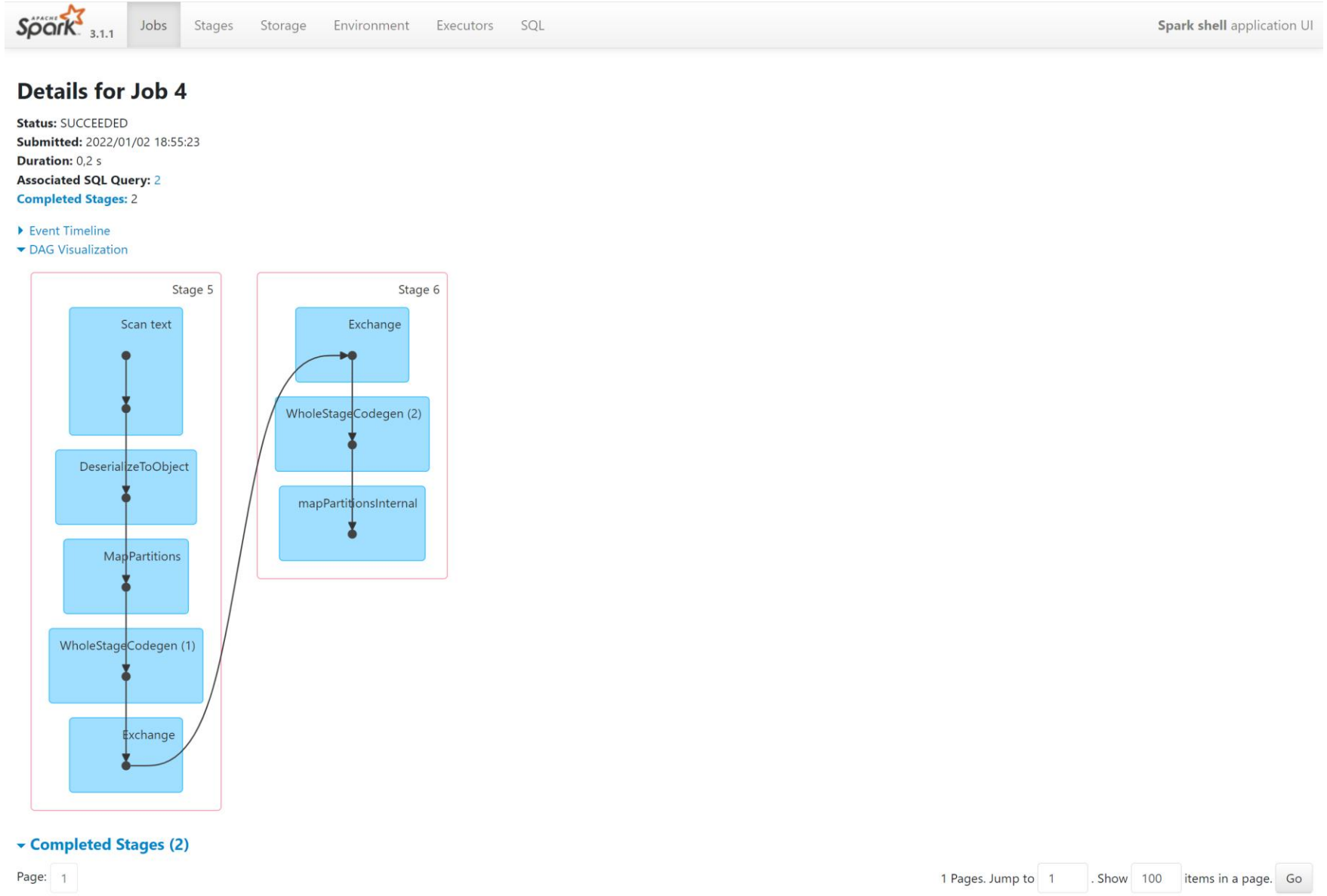
http://localhost:4040



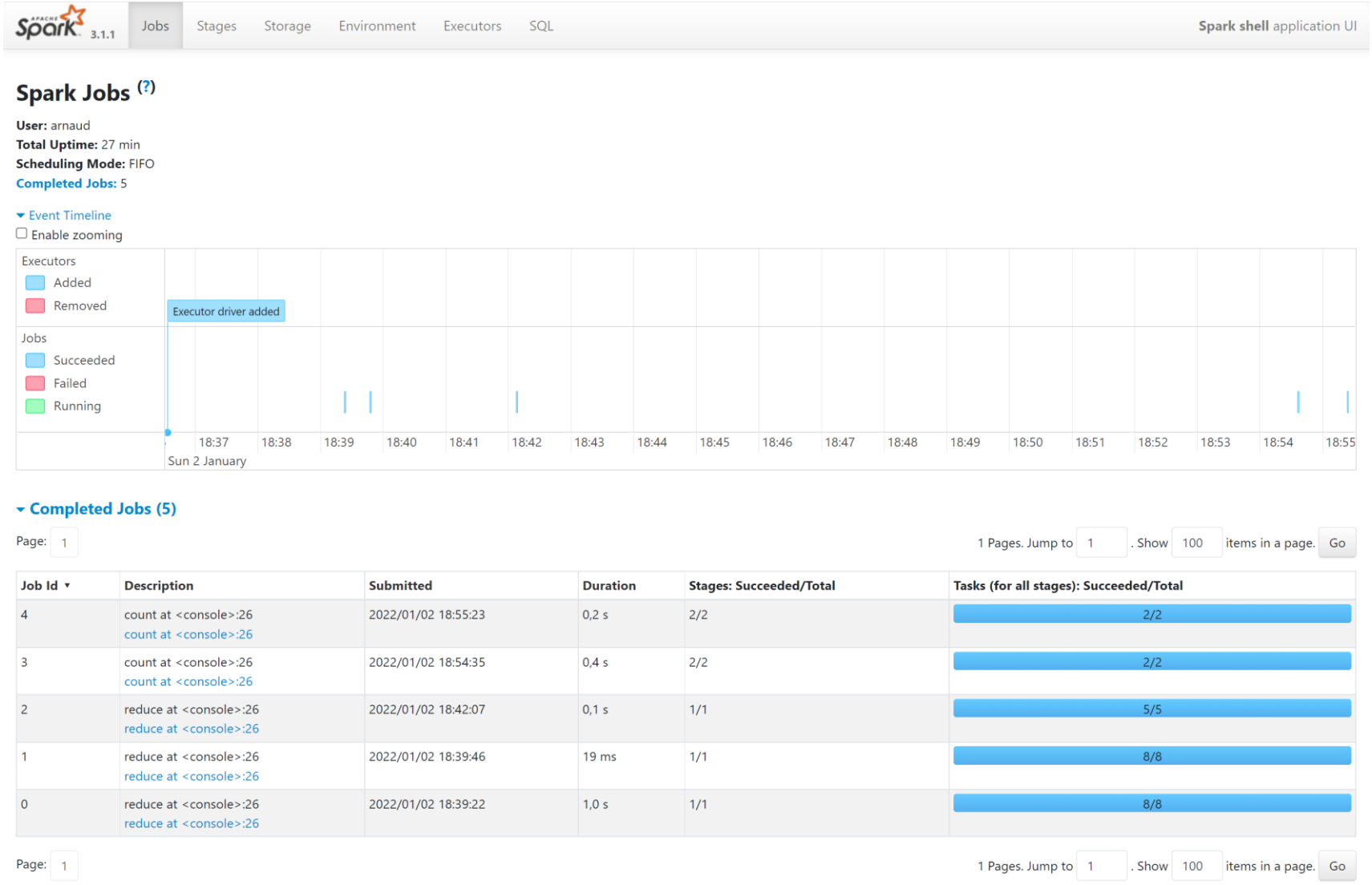
spark-ui [1/6]: Jobs



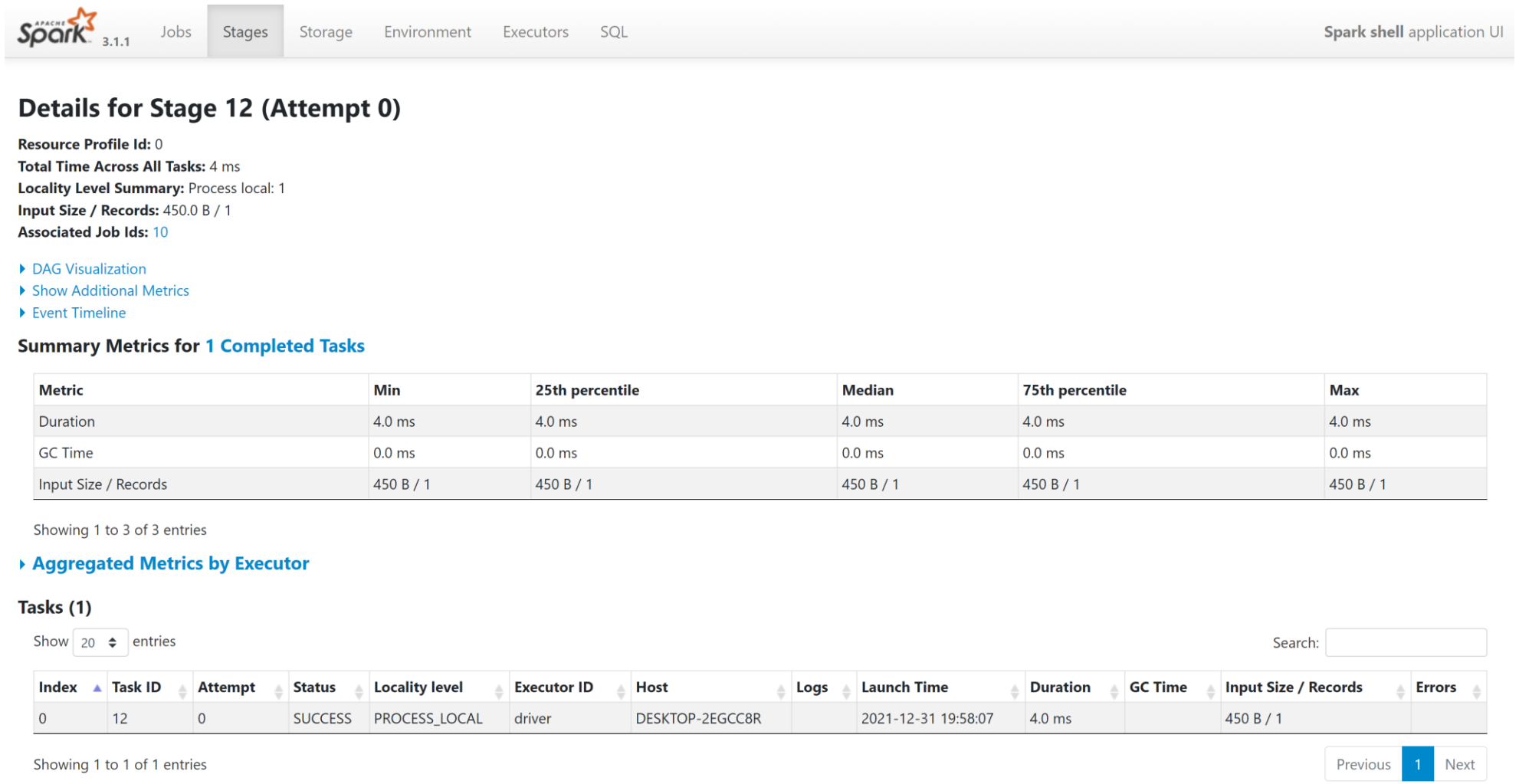
Spark-ui [1/6]: Job details DAG



Spark-ui [1/6] detail Job timeline + statistics



Spark-ui [2/6]: detail Job > Stage > Task



Spark-ui [3/6] Storage (cache, persist)

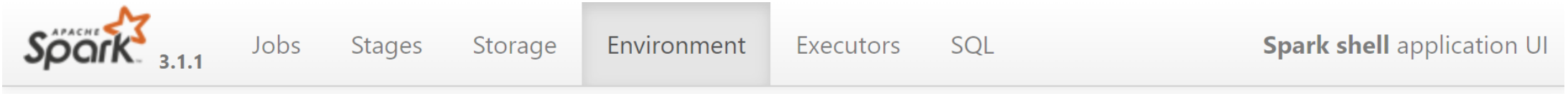
Spark 3.1.1 Jobs Stages Storage Environment Executors SQL **Spark shell** application UI

Storage

RDDs

ID	RDD Name	Storage Level	Cached Partitions	Fraction Cached	Size in Memory	Size on Disk
28	*(1) SerializeFromObject [staticinvoke(class org.apache.spark.unsafe.types.UTF8String, StringType, fromString, input[0, java.lang.String, true], true, false) AS value#30] +- MapPartitions org.apache.spark.sql.Dataset$$Lambda$4450/17435982@68a3d4, obj#29: java.lang.String +- DeserializeToObject value#21.toString, obj#28: java.lang.String +- FileScan text [value#21] Batched: false, DataFilters: [], Format: Text, Location: InMemoryFileIndex[file:/c:/data/loremIpsum.txt], PartitionFilters: [], PushedFilters: [], ReadSchema: struct<value:string>	Disk Memory Deserialized 1x Replicated	1	100%	952.0 B	0.0 B
40	FileScan text [value#21] Batched: false, DataFilters: [], Format: Text, Location: InMemoryFileIndex[file:/c:/data/loremIpsum.txt], PartitionFilters: [], PushedFilters: [], ReadSchema: struct<value:string>	Disk Memory Deserialized 1x Replicated	1	100%	960.0 B	0.0 B

Spark-ui [4/6] Environment: jvm+spark+hadoop+..



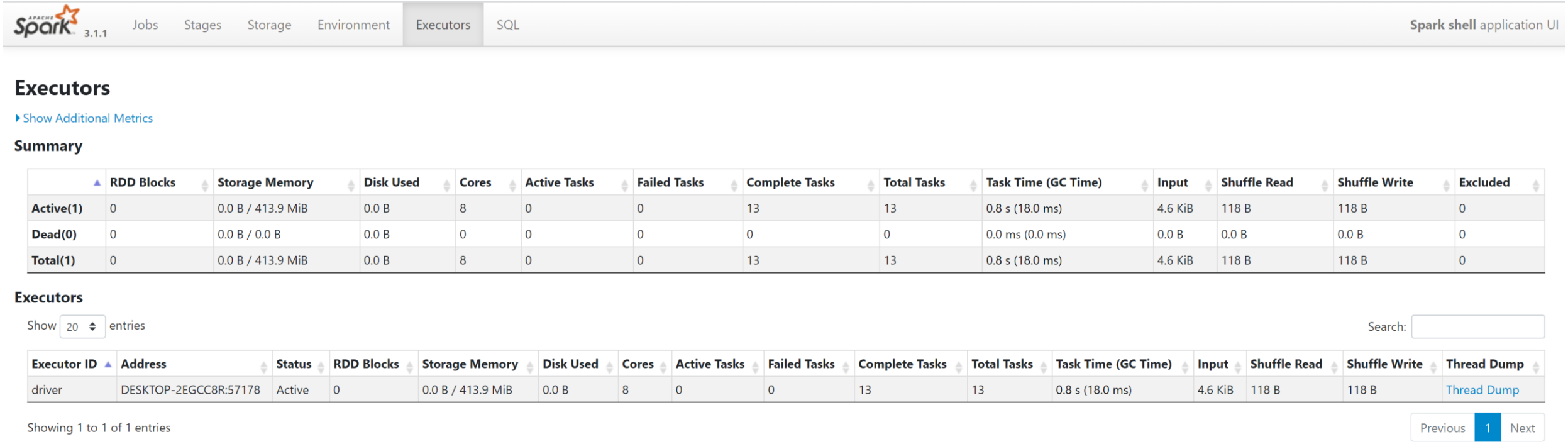
Environment

- Runtime Information
- Spark Properties
- Resource Profiles
- Hadoop Properties
- System Properties
- Classpath Entries

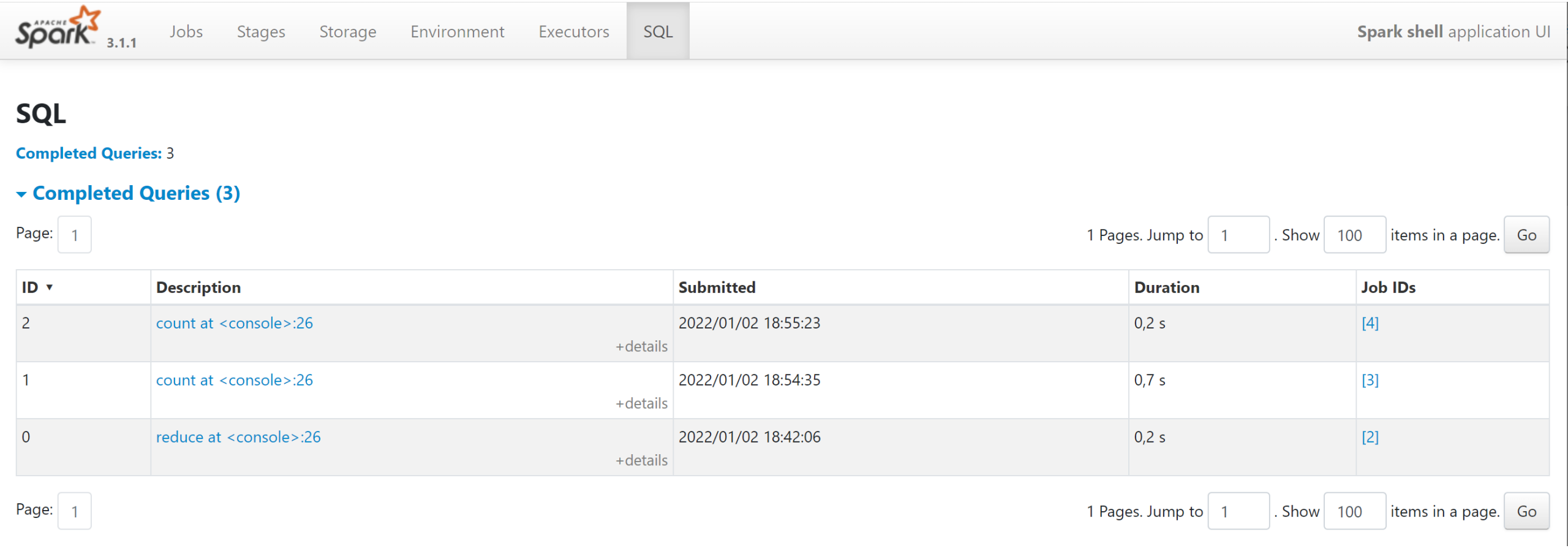
▾ Spark Properties

Name	Value
spark.app.id	local-1640975191762
spark.app.name	Spark shell
spark.app.startTime	1640975189843
spark.driver.host	DESKTOP-2EGCC8R
spark.driver.port	57122
spark.executor.id	driver
spark.home	C:\apps\hadoop\spark-3.1.1
spark.jars	
spark.master	local[*]
spark.repl.class.outputDir	C:\Users\arnaud\AppData\Local\Temp\spark-169a28ae-4479-4585-ab47-1e354d2d1347\repl-ff2cda14-38ec-4b1e-9c63-c4f02e84bf14
spark.repl.class.uri	spark://DESKTOP-2EGCC8R:57122/classes
spark.scheduler.mode	FIFO
spark.sql.catalogImplementation	hive
spark.submit.deployMode	client
spark.submit.pyFiles	
spark.ui.showConsoleProgress	true

Spark-ui [5/6]: Executors / Driver



Spark-ui [6/6] SQL



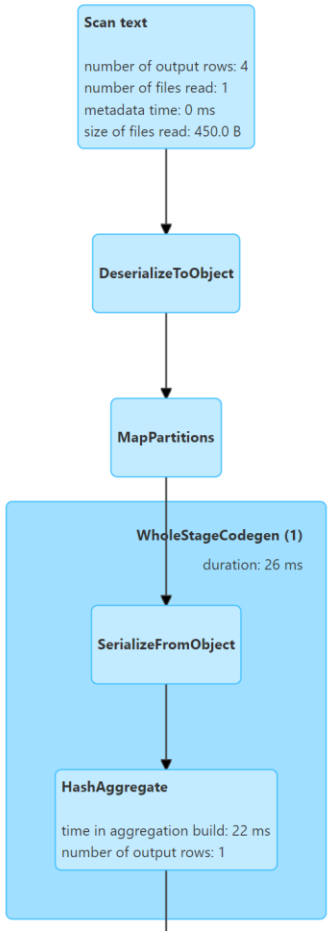
Spark-ui [6/6] Sql detail for query

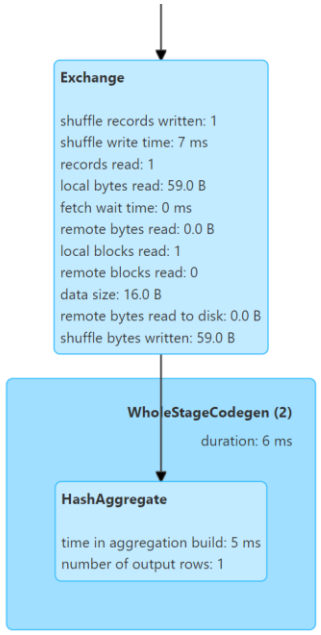
Spark 3.1.1 | Jobs | Stages | Storage | Environment | Executors | SQL

Details for Query 2

Submitted Time: 2022/01/02 18:55:23
Duration: 0,2 s
Succeeded Jobs: 4

☐ Show the Stage ID and Task ID that corresponds to the max metric





▾ Details

```
== Physical Plan ==
* HashAggregate (7)
+- Exchange (6)
   +- * HashAggregate (5)
      +- * SerializeFromObject (4)
         +- MapPartitions (3)
            +- DeserializeToObject (2)
               +- Scan text  (1)


(1) Scan text
Output [1]: [value#21]
Batched: false
Location: InMemoryFileIndex [file:/c:/data/loremIpsum.txt]
ReadSchema: struct<value:string>

(2) DeserializeToObject
Input [1]: [value#21]
Arguments: value#21.toString, obj#28: java.lang.String

(3) MapPartitions
Input [1]: [obj#28]
Arguments: org.apache.spark.sql.Dataset$$Lambda$4450/17435982@68a3d4, obj#29: java.lang.String
```

```
(4) SerializeFromObject [codegen id : 1]
Input [1]: [obj#29]

(5) HashAggregate [codegen id : 1]
Input: []
Keys: []
Functions [1]: [partial_count(1)]
Aggregate Attributes [1]: [count#35L]
Results [1]: [count#36L]

(6) Exchange
Input [1]: [count#36L]
Arguments: SinglePartition, ENSURE_REQUIREMENTS, [id=#55]

(7) HashAggregate [codegen id : 2]
Input [1]: [count#36L]
Keys: []
Functions [1]: [count(1)]
Aggregate Attributes [1]: [count(1)#32L]
Results [1]: [count(1)#32L AS count#33L]
```

Spark Internals

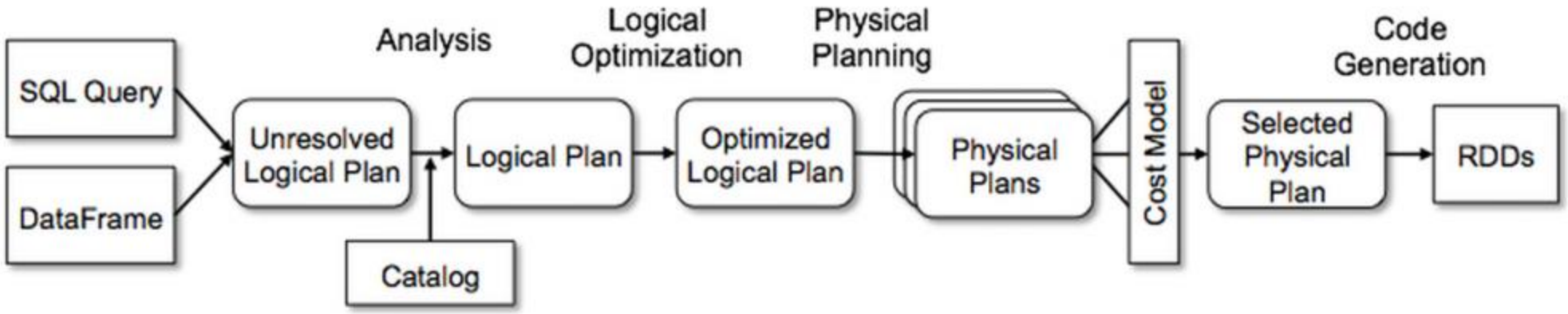
Logical/PhysicalPlan

Jobs - Stage - Tasks

Sql -> parsed AST -> Dataset + expression API

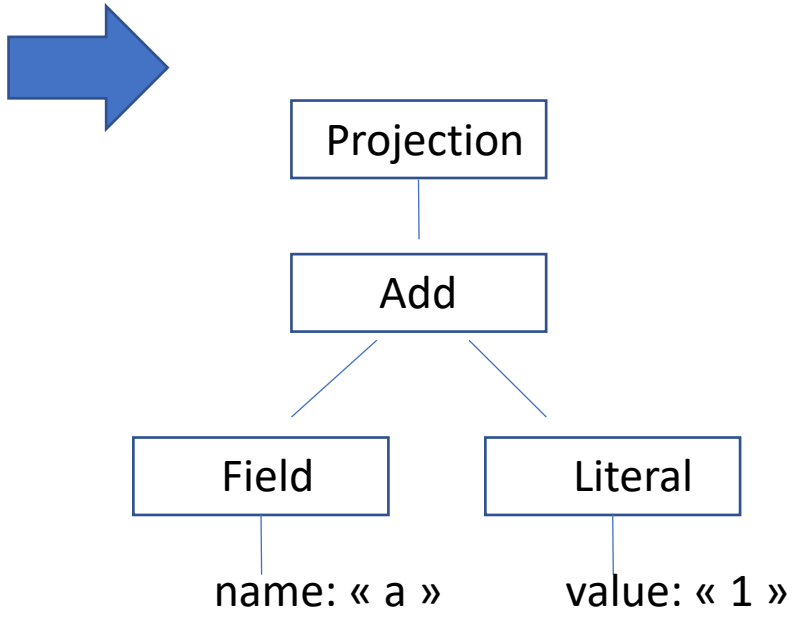
Optimisation: re-write
from LogicalPlan to PhysicalPlan

SQL / DataSet -> AST -> Catalyst -> Code + RDD



Compiler chain: Parser -> AST -> ..

Sql> select **(a+1)** from …




```scala
case class Add(left: Expression, right: Expression) extends BinaryArithmetic {
  override def symbol: String = "+"

  lazy val numeric = dataType match {
    case n: NumericType => n.numeric.asInstanceOf[Numeric[Any]]
    case other => sys.error(s"Type $other does not support numeric operations")
  }

  override def eval(input: Row): Any = {
    val evalE1 = left.eval(input)
    if(evalE1 == null) {
      null
    } else {
      val evalE2 = right.eval(input)
      if (evalE2 == null) {
        null
      } else {
        numeric.plus(evalE1, evalE2)
      }
    }
  }
}
```

Catalyst AST -> CodeGenerator

apache / spark Public

Watch 2.1k | Fork 25k | Star 31.7k

Code | Pull requests 241 | Actions | Projects | Security | Insights

fedbfc7074 spark / sql / catalyst / src / main / scala / org / apache / spark / sql / catalyst / expressions / codegen / CodeGenerator.scala

Go to file ...

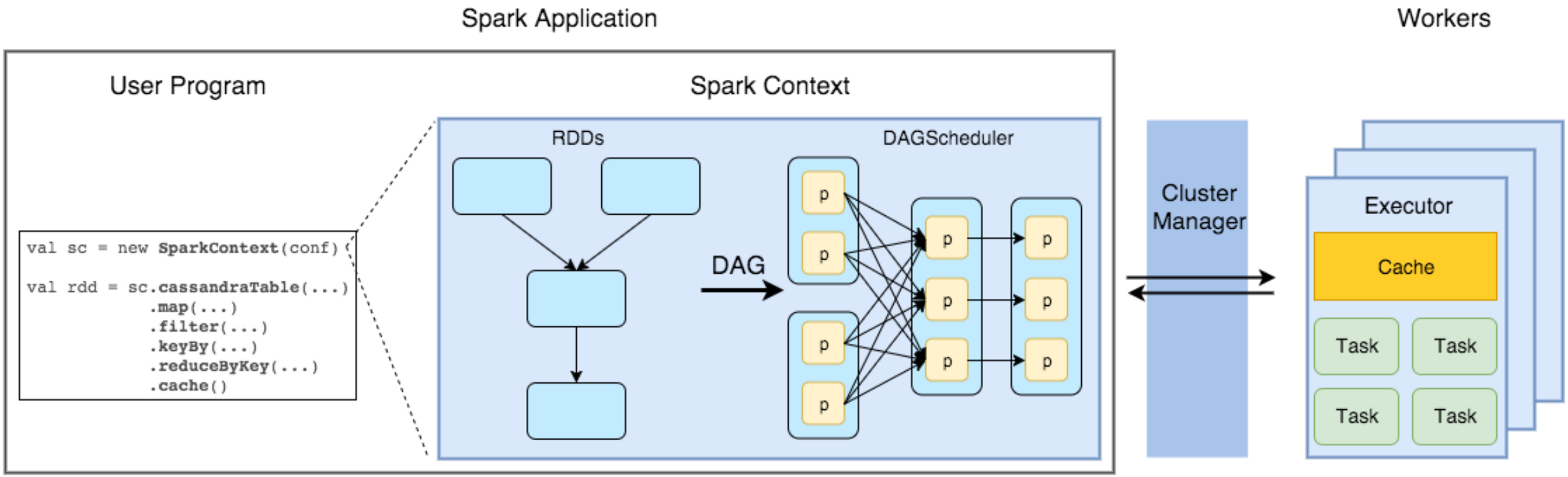
```
33  /**
34   * A base class for generators of byte code to perform expression evaluation.  Includes a set of
35   * helpers for referring to Catalyst types and building trees that perform evaluation of individual
36   * expressions.
37   */
38  abstract class CodeGenerator[InType <: AnyRef, OutType <: AnyRef] extends Logging {
```

```
363         case Add(e1, e2) =>      (e1, e2) evaluate { case (eval1, eval2) => q"$eval1 + $eval2" }
364         case Subtract(e1, e2) => (e1, e2) evaluate { case (eval1, eval2) => q"$eval1 - $eval2" }
365         case Multiply(e1, e2) => (e1, e2) evaluate { case (eval1, eval2) => q"$eval1 * $eval2" }
```

```
640     }
641   }
```

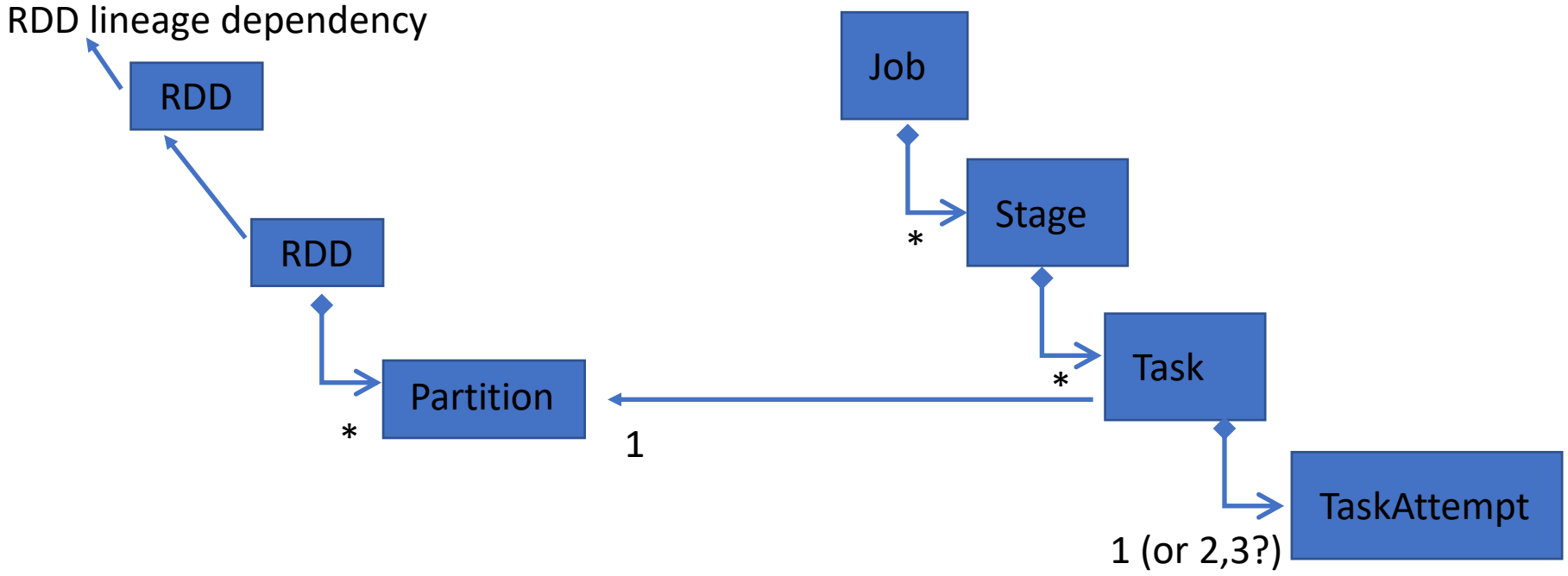
Runtime Execution:

Dataset -> RDD -> Dag Scheduler
-> Jobs,Stages,Tasks

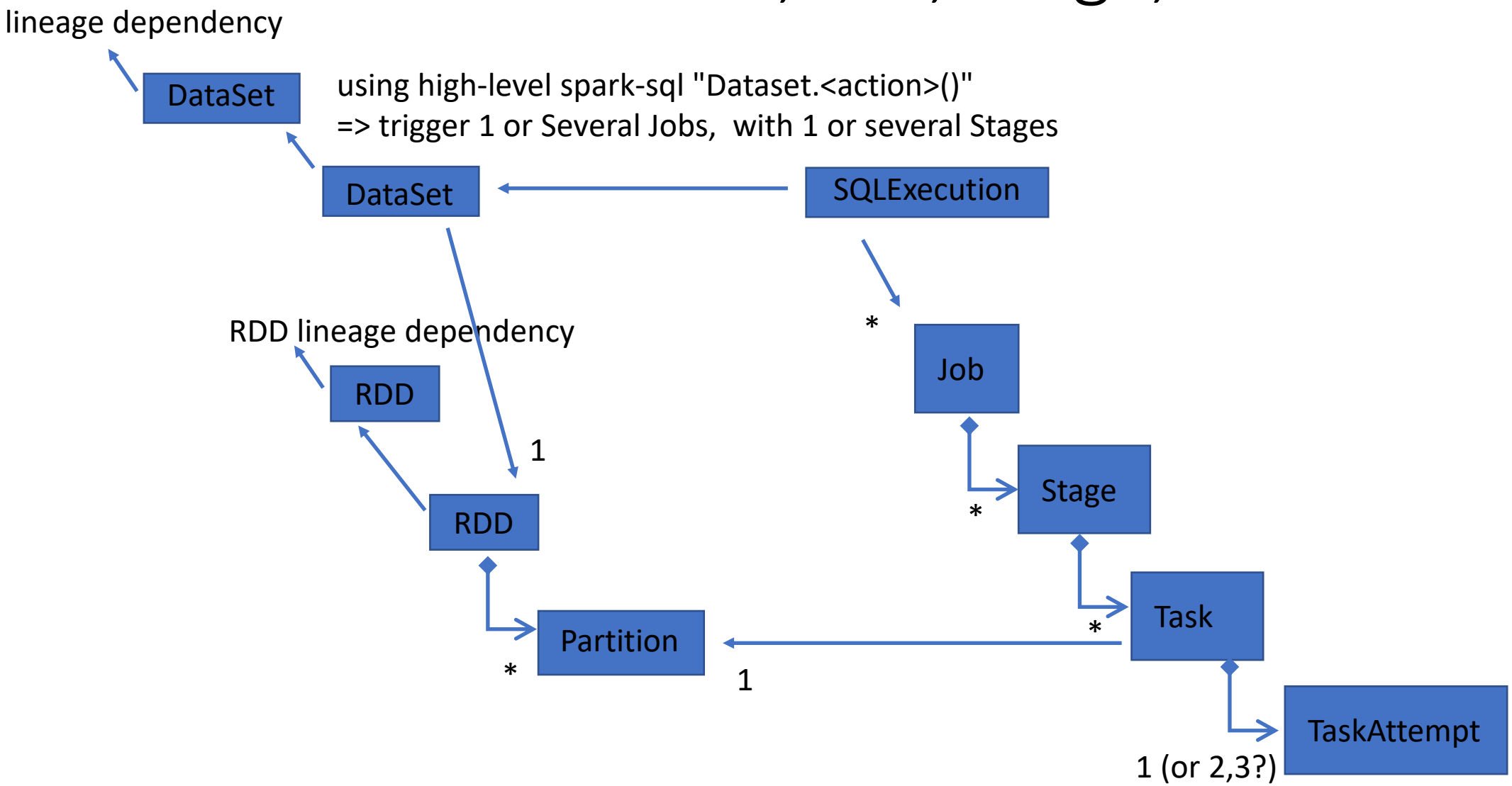

Spark Application
Workers
User Program
Spark Context
val sc = new SparkContext(conf)
val rdd = sc.cassandraTable(...)
.map(...)
.filter(...)
.keyBy(...)
.reduceByKey(...)
.cache()
RDDs
DAGScheduler
DAG
p
Cluster Manager
Executor
Cache
Task
Task
Task
Task

Spark-core Internal Concepts…
RDD, Job, Stage, Task

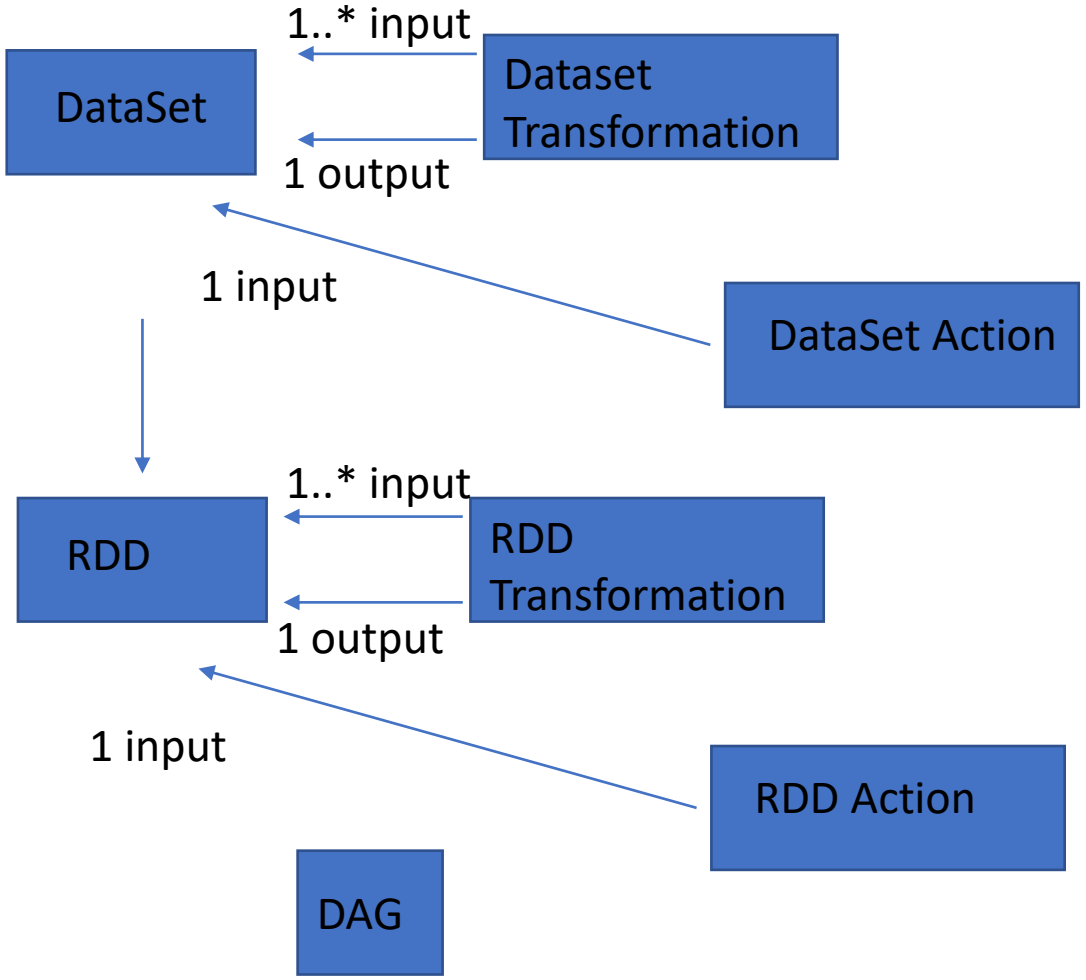
using only low-level spark-core "RDD.eval()"
=> trigger 1 Job, with 1 or several Stages

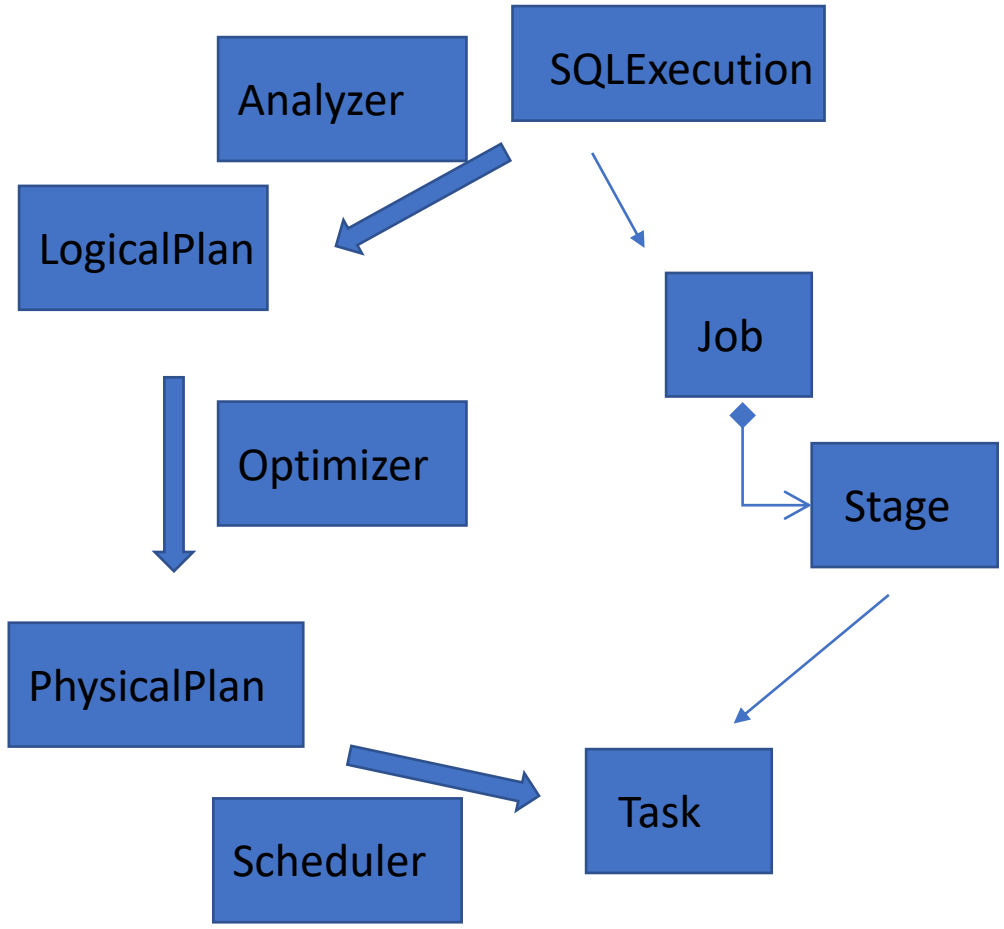


Spark-sql Internal Concepts… SQLExecution, Job, Stage, Task



Spark Internal Concepts…
SQL, Job, Stage, Task

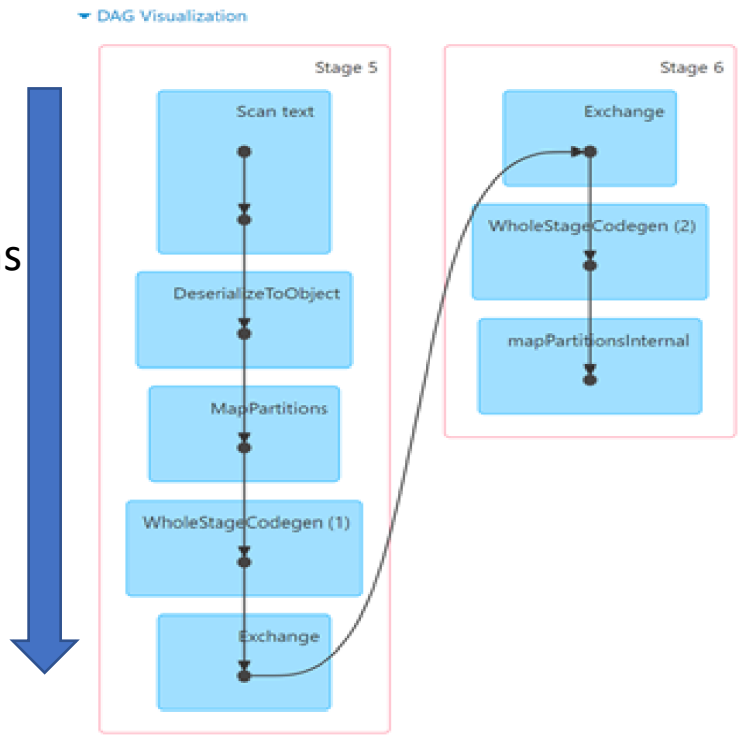




DAG Job = list of Stages
each Stage = N "narrow" transformations
between Stages = Exchange Shuffle ("Wide" transfo.)

1 stage =
Successive **Narrow** transformations

Bytecode generated
… executed within same
Thread/Process



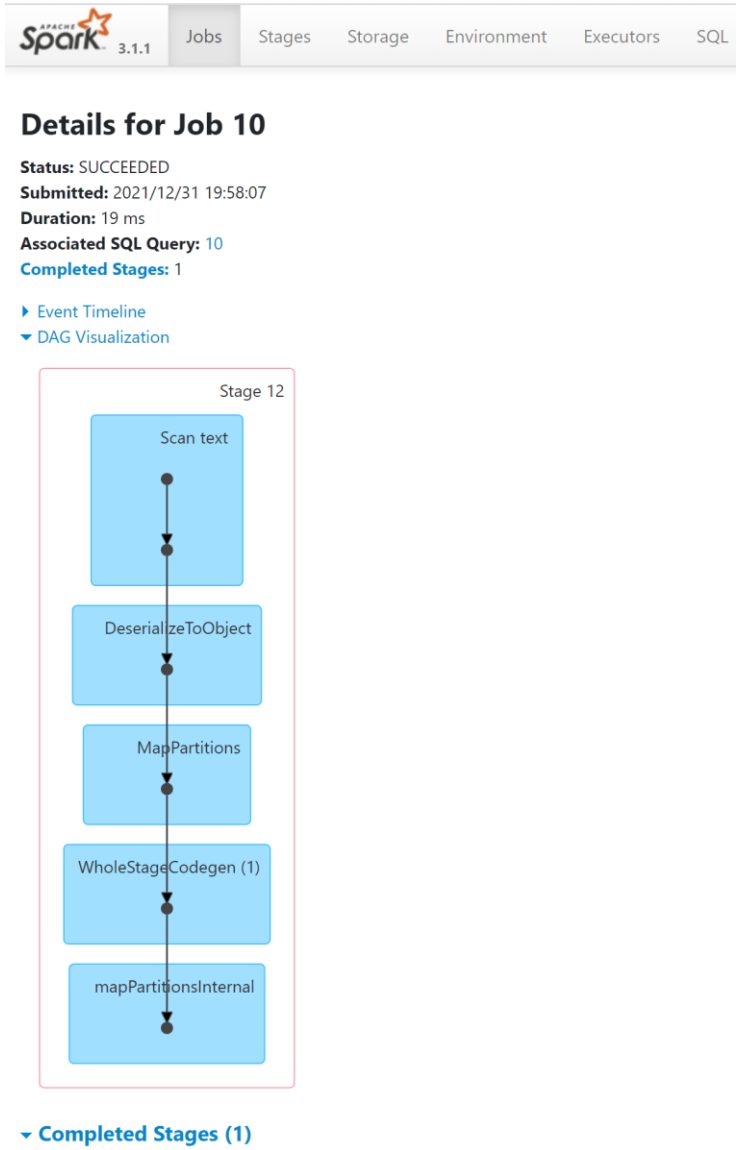
Job =
Full DAG of Stages

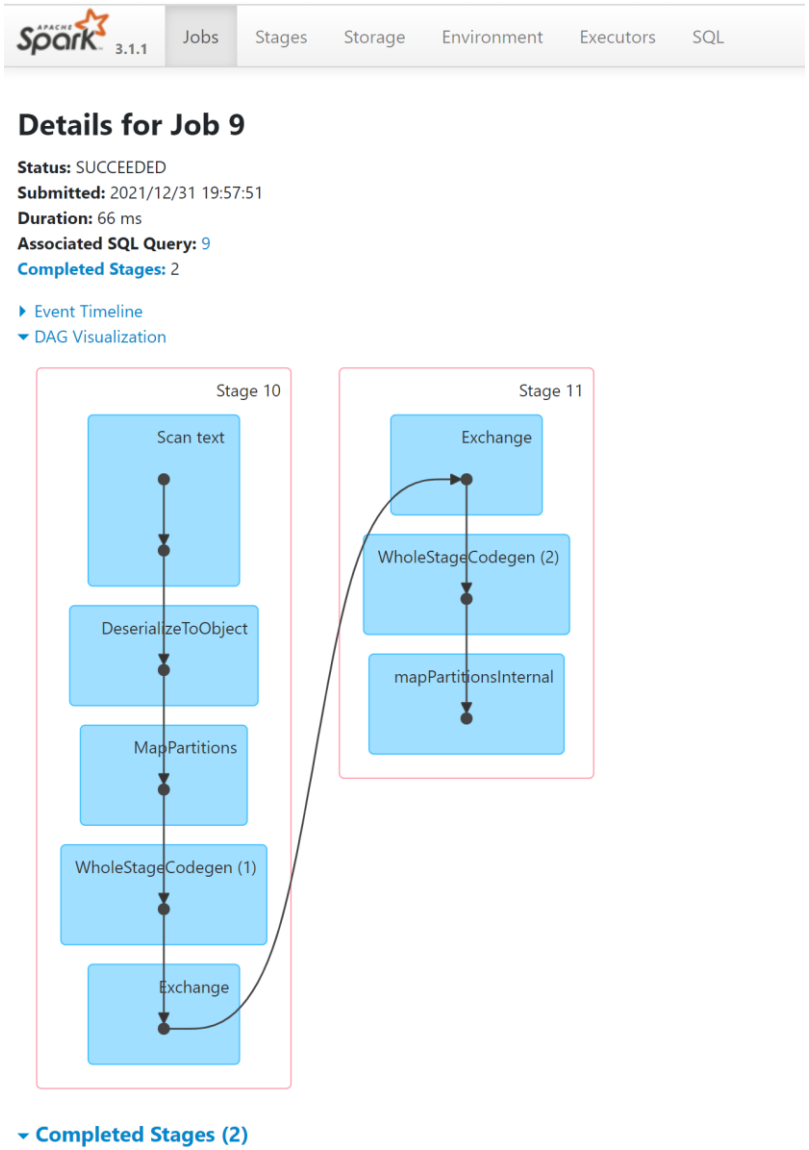
Exchange
Output

Exchange
Input

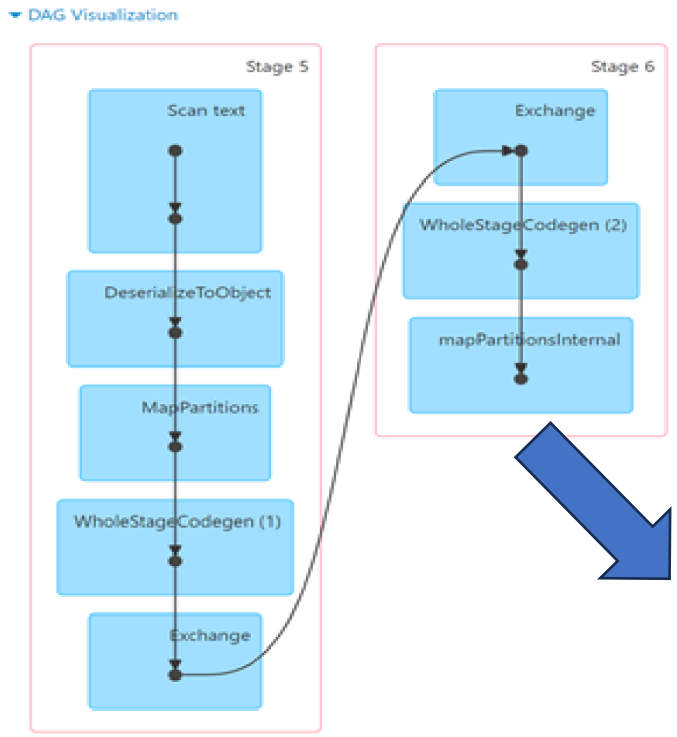
Wide transformation … separation between successive **Stages**

example Word Count DAGs
simple -vs- repartition()

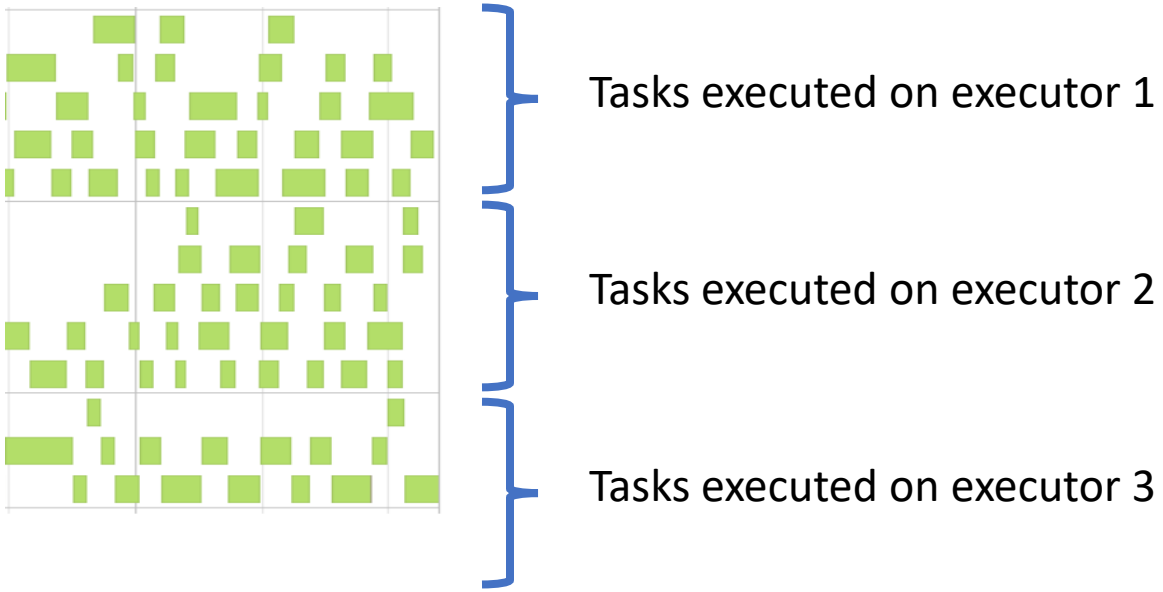




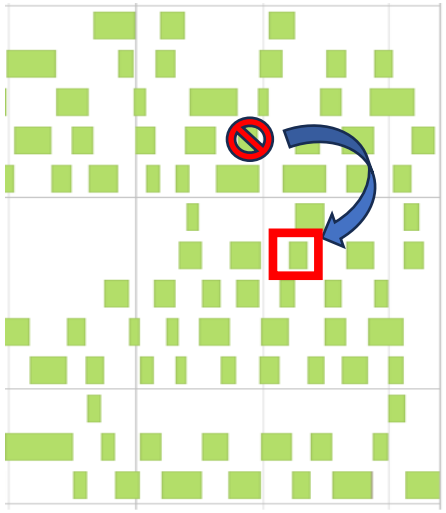
Stage = list of Tasks : 1 per Partition
… each scheduled on an executor



Tasks "**eventTimeline**" for stage



Tasks may fail, Retry individually

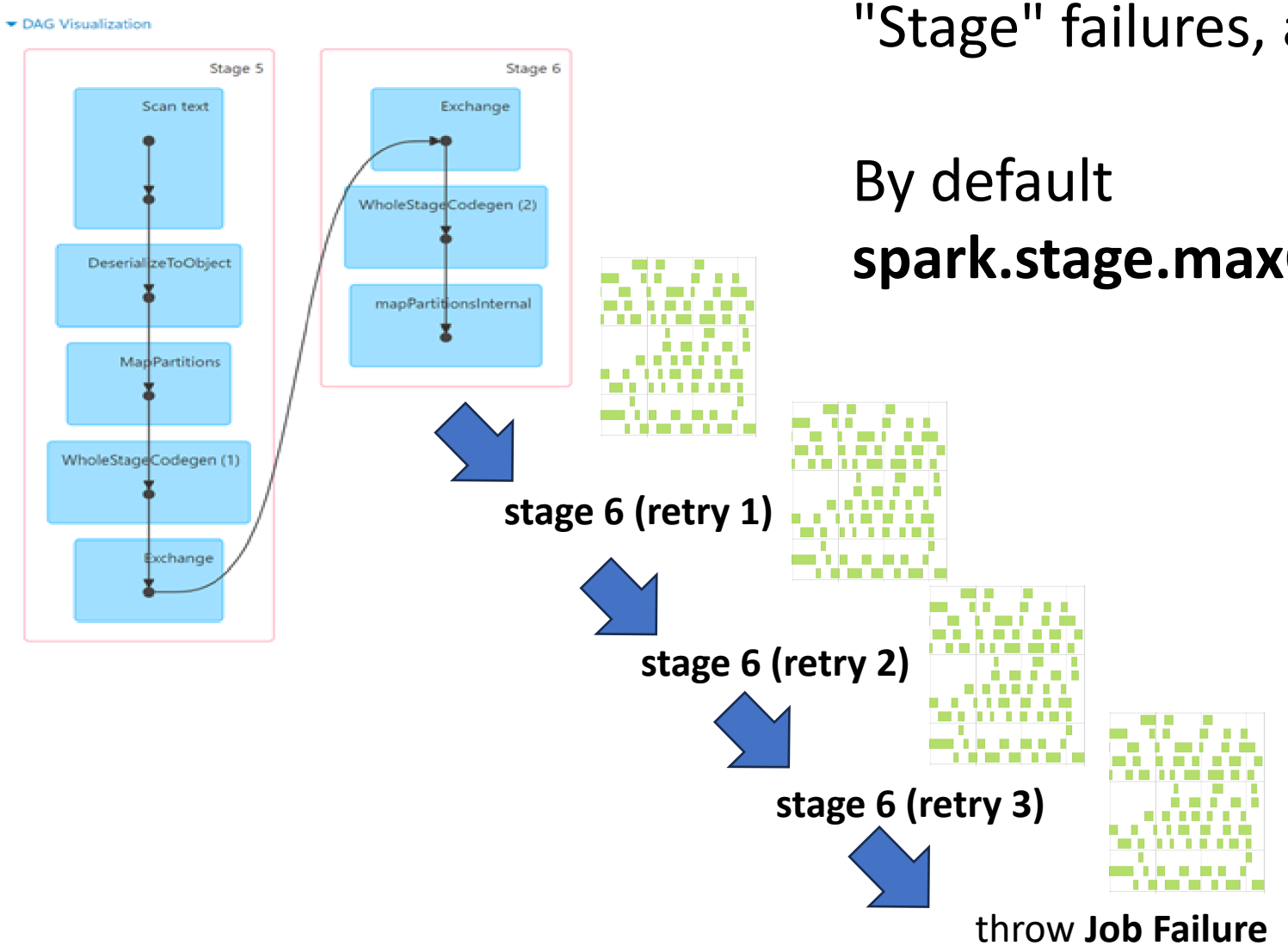


Tasks errors are automatically retryed (rescheduled)

By default,

spark.task.maxFailures = 4

Stage Retry



"Stage" failures, are also retryed

By default

spark.stage.maxConsecutiveAttempts = 4

Spark History Server

How to analyze a finished spark-submit App?

Typical use-case :

most daily batches run once per day, late or during night

How to analyze performances, but OFFLINE ?
(process finished => not able to connect to « http://localhost:4040 »)

How to compare performances with older applications

eventLog Files, disabled by default

Extract of file ${SPARK_HOME} / conf / spark-default.conf.template

```
# spark.eventLog.enabled     true
# spark.eventLog.dir         hdfs://namenode:8021/directory
…
```

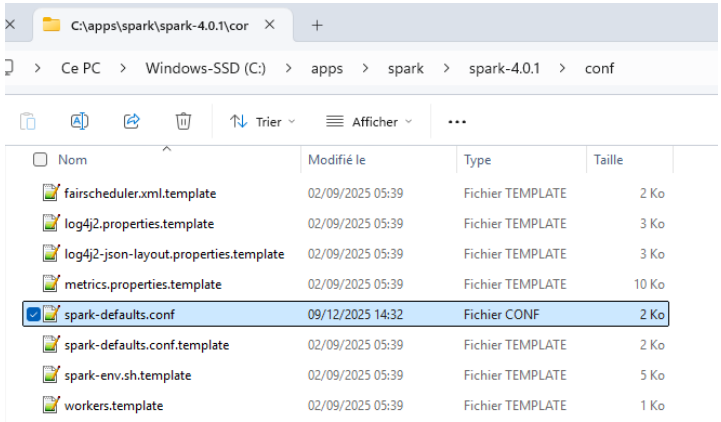
By default, eventLog Files are disabled

(no « spark-default.conf » file, and in the template it is commented « # »)

Notice: eventLog Files are DIFFERENT FROM   log4j stdout files
By default, there is also NO log file (only WARN message are written to console)

Enabling eventLog Files

Copy file spark-defaults.conf.template to spark-defaults.conf



+ Edit to uncomment and configure log dir

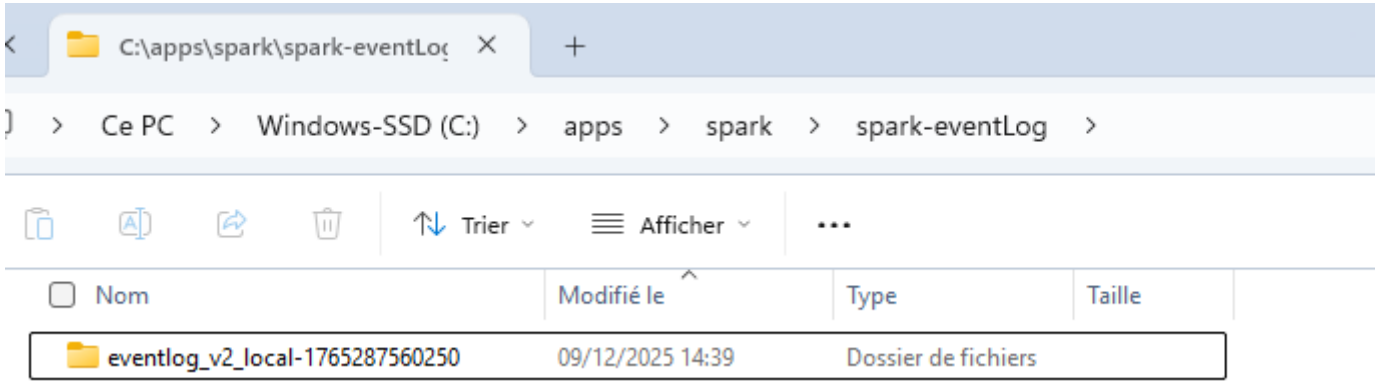
spark.eventLog.enabled true
spark.eventLog.dir file:///C:/apps/spark/spark-eventLog

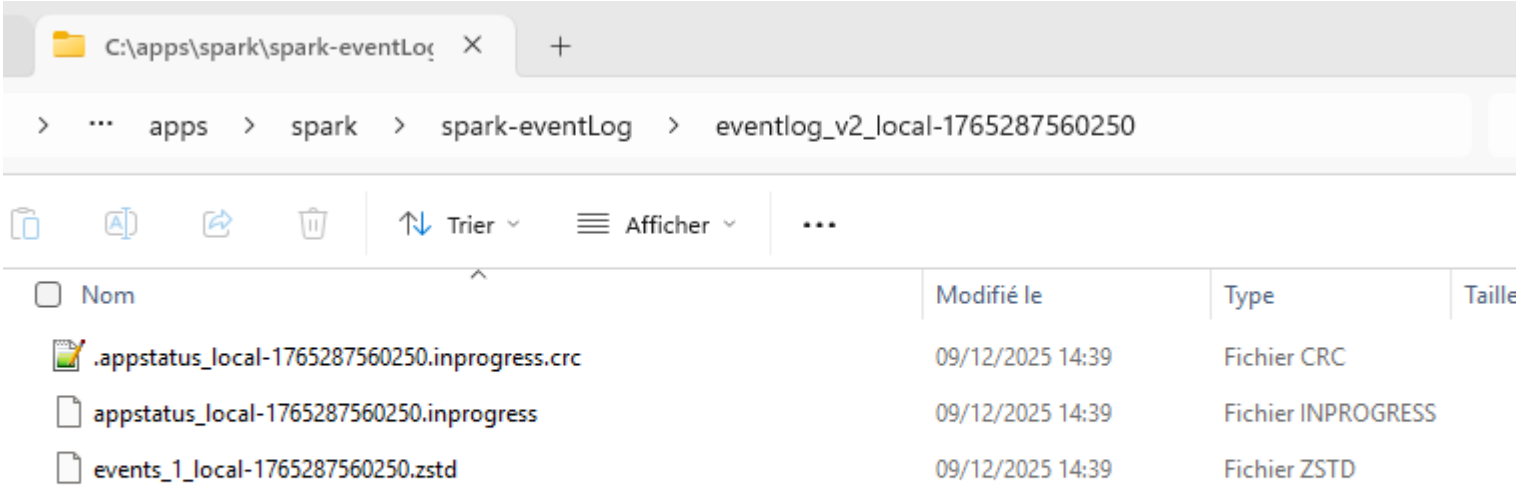
also edit for HistoryServer
spark.history.fs.logDirectory file:///C:/apps/spark/spark-eventLog

Or also explicit launch:

```
spark-shell \
    --conf spark.evenLog.enabled=true  \
    --conf spark.eventLog.dir=file:///C:/apps/spark/spark-eventLog
```

Launching spark => 1 Dir + N x written Files





Uncompressing EventLog Files, internal Check

```
arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ ls -l
total 96
-rw-rw----+ 1 arnaud Aucun     0 Dec  9 14:39 appstatus_local-1765287560250.inprogress
-rw-rw----+ 1 arnaud Aucun 64997 Dec  9 14:42 events_1_local-1765287560250.zstd

arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ cat events_1_local-1765287560250.zstd | zstd -d > events_1_local-1765287560250.ndjson

arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ ls -l
total 836
-rw-rw----+ 1 arnaud Aucun      0 Dec  9 14:39 appstatus_local-1765287560250.inprogress
-rw-r--r--+ 1 arnaud Aucun 753747 Dec  9 14:45 events_1_local-1765287560250.ndjson
-rw-rw----+ 1 arnaud Aucun  64997 Dec  9 14:42 events_1_local-1765287560250.zstd

arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ less events_1_local-1765287560250.ndjson

arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ head -4 events_1_local-1765287560250.ndjson
{"Event":"SparkListenerLogStart","Spark Version":"4.0.1"}
{"Event":"SparkListenerResourceProfileAdded","Resource Profile Id":0,"Executor Resource Requests":{"cores":{"Resource Name":"cores","Amount":1,"Discovery
 Script":"","Vendor":""},"memory":{"Resource Name":"memory","Amount":1024,"Discovery Script":"","Vendor":""},"offHeap":{"Resource Name":"offHeap","Amount
":0,"Discovery Script":"","Vendor":""}},"Task Resource Requests":{"cpus":{"Resource Name":"cpus","Amount":1.0}}}
{"Event":"SparkListenerExecutorAdded","Timestamp":1765287560391,"Executor ID":"driver","Executor Info":{"Host":"DesktopArnaud","Total Cores":8,"Log Urls"
:{},"Attributes":{},"Resources":{},"Resource Profile Id":0}}
{"Event":"SparkListenerBlockManagerAdded","Block Manager ID":{"Executor ID":"driver","Host":"DesktopArnaud","Port":64835},"Maximum Memory":3032481792,"Ti
mestamp":1765287560476,"Maximum Onheap Memory":3032481792,"Maximum Offheap Memory":0}
```

eventLog Files Source Details…

eventLog = ND-JSON file containing internal SparkListener Bus events

https://github.com/apache/spark/blob/master/core/src/main/scala/org/apache/spark/scheduler/SparkListener.scala

spark / core / src / main / scala / org / apache / spark / scheduler / SparkListener.scala

Code | Blame | 588 lines (476 loc) · 18.4 KB

```scala
package org.apache.spark.scheduler

import java.util.Properties
import javax.annotation.Nullable

import scala.collection.Map

import org.apache.spark.TaskEndReason
import org.apache.spark.annotation.{DeveloperApi, Since}
import org.apache.spark.executor.{ExecutorMetrics, TaskMetrics}
import org.apache.spark.resource.ResourceProfile
import org.apache.spark.scheduler.cluster.ExecutorInfo
import org.apache.spark.storage.{BlockManagerId, BlockUpdatedInfo}

@DeveloperApi
case class SparkListenerStageSubmitted(stageInfo: StageInfo, properties: Properties = null)
  extends SparkListenerEvent

@DeveloperApi
case class SparkListenerStageCompleted(stageInfo: StageInfo) extends SparkListenerEvent

@DeveloperApi
case class SparkListenerTaskStart(stageId: Int, stageAttemptId: Int, taskInfo: TaskInfo)
  extends SparkListenerEvent

@DeveloperApi
case class SparkListenerTaskGettingResult(taskInfo: TaskInfo) extends SparkListenerEvent

@DeveloperApi
case class SparkListenerSpeculativeTaskSubmitted(
    stageId: Int,
    stageAttemptId: Int = 0)
  extends SparkListenerEvent {
```

https://github.com/apache/spark/blob/master/core/src/main/scala/org/apache/spark/util/JsonProtocol.scala

spark / core / src / main / scala / org / apache / spark / util / JsonProtocol.scala ↑ Top

Code | Blame | 1705 lines (1568 loc) · 75.4 KB | Raw

```scala
  val defaultOptions: JsonProtocolOptions = new JsonProtocolOptions(new SparkConf(false))
  def sparkEventToJsonString(event: SparkListenerEvent, options: JsonProtocolOptions): String = {
    ...
  }

  def writeSparkEventToJson(
      event: SparkListenerEvent,
      g: JsonGenerator,
      options: JsonProtocolOptions): Unit = {
    event match {
      case stageSubmitted: SparkListenerStageSubmitted =>
        stageSubmittedToJson(stageSubmitted, g, options)
      case stageCompleted: SparkListenerStageCompleted =>
        stageCompletedToJson(stageCompleted, g, options)
      case taskStart: SparkListenerTaskStart =>
        taskStartToJson(taskStart, g, options)
      case taskGettingResult: SparkListenerTaskGettingResult =>
        taskGettingResultToJson(taskGettingResult, g, options)
      case taskEnd: SparkListenerTaskEnd =>
        taskEndToJson(taskEnd, g, options)
      case jobStart: SparkListenerJobStart =>
        jobStartToJson(jobStart, g, options)
      case jobEnd: SparkListenerJobEnd =>
        jobEndToJson(jobEnd, g)
      case environmentUpdate: SparkListenerEnvironmentUpdate =>
        environmentUpdateToJson(environmentUpdate, g)
      case blockManagerAdded: SparkListenerBlockManagerAdded =>
        blockManagerAddedToJson(blockManagerAdded, g)
      case blockManagerRemoved: SparkListenerBlockManagerRemoved =>
        blockManagerRemovedToJson(blockManagerRemoved, g)
      case unpersistRDD: SparkListenerUnpersistRDD =>
        unpersistRDDToJson(unpersistRDD, g)
      case applicationStart: SparkListenerApplicationStart =>
```

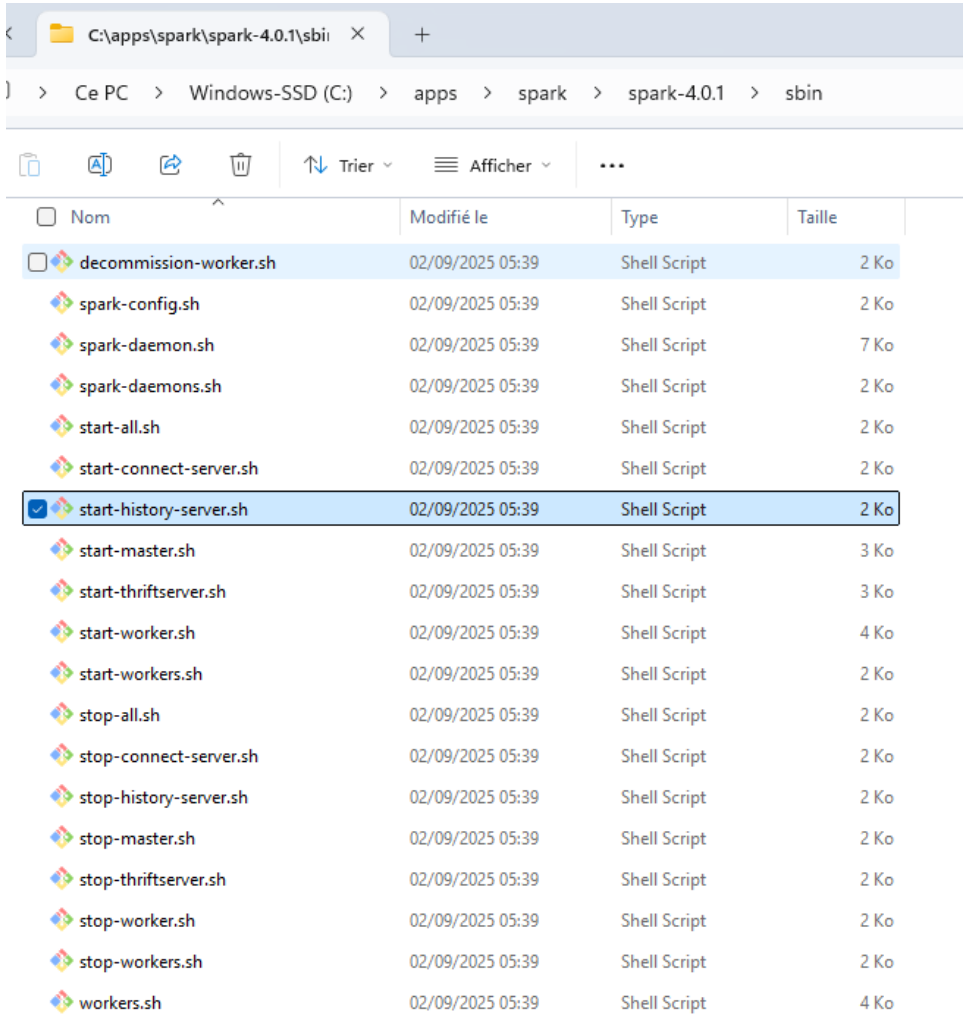
Running Spark History Server

The History Server, is the same code that Spark UI,
but running as a Web Server to view old eventLog Files

To run it …

${SPARK_HOME}/sbin/start-history-server.sh

Notice:
- in « sbin » directory, not in « bin » (so not in your PATH)
- Only « .sh » bash script, no windows « .cmd »



Spark History Server …

sbin/spark-history-server.sh

➔ does not work on windows, even with cmder + bash

➔ is almost equivalent to …

spark-class.cmd org.apache.spark.deploy.history.HistoryServer

➔ start but expect different conf

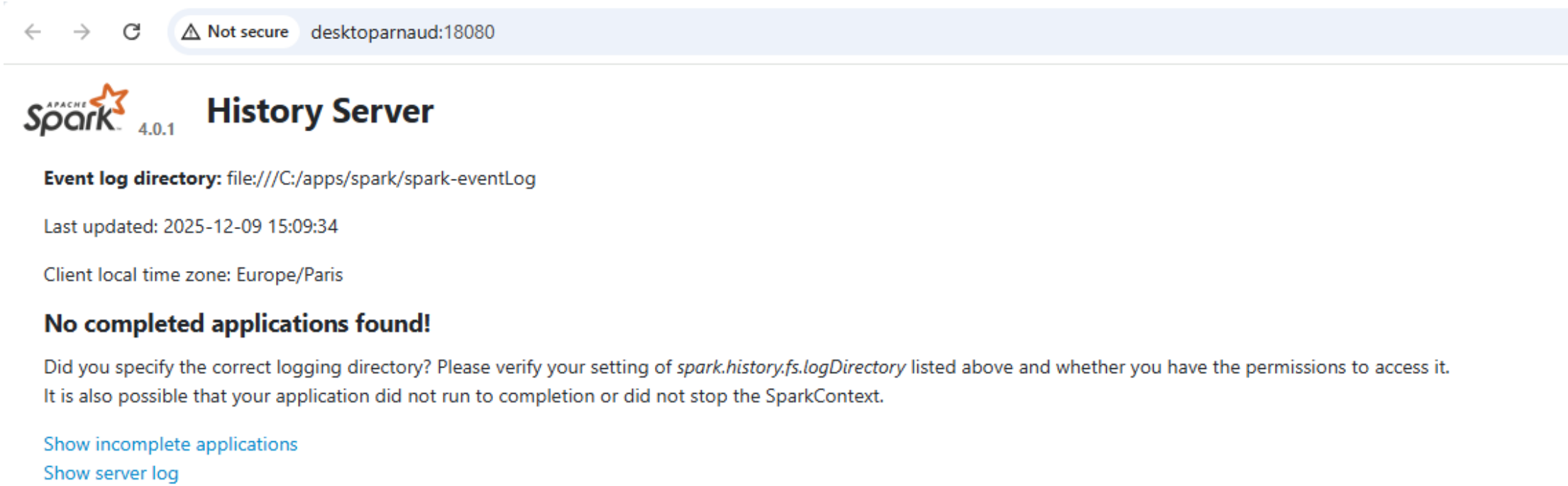
spark.eventLog.dir file:///C:/apps/spark/spark-eventLog

spark.history.fs.logDirectory file:///C:/apps/spark/spark-eventLog

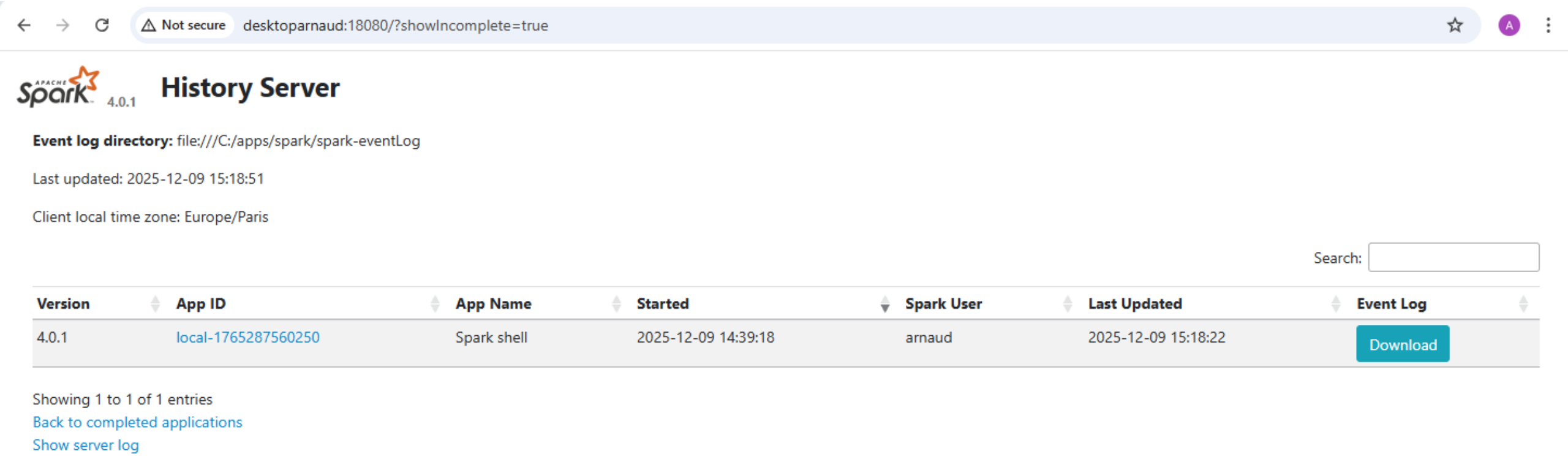
```
25/12/09 15:08:02 INFO Utils: Successfully started service 'HistoryServerUI' on port 18080.
25/12/09 15:08:02 INFO HistoryServer: Bound HistoryServer to 0.0.0.0, and started at http://DesktopArnaud:18080
Exception in thread "main" java.io.FileNotFoundException: Log directory specified does not exist: file:/tmp/spark-events Did you configure
the correct one through spark.history.fs.logDirectory?
        at org.apache.spark.deploy.history.FsHistoryProvider.startPolling(FsHistoryProvider.scala:276)
        at org.apache.spark.deploy.history.FsHistoryProvider.initialize(FsHistoryProvider.scala:221)
```

http://localhost:18080

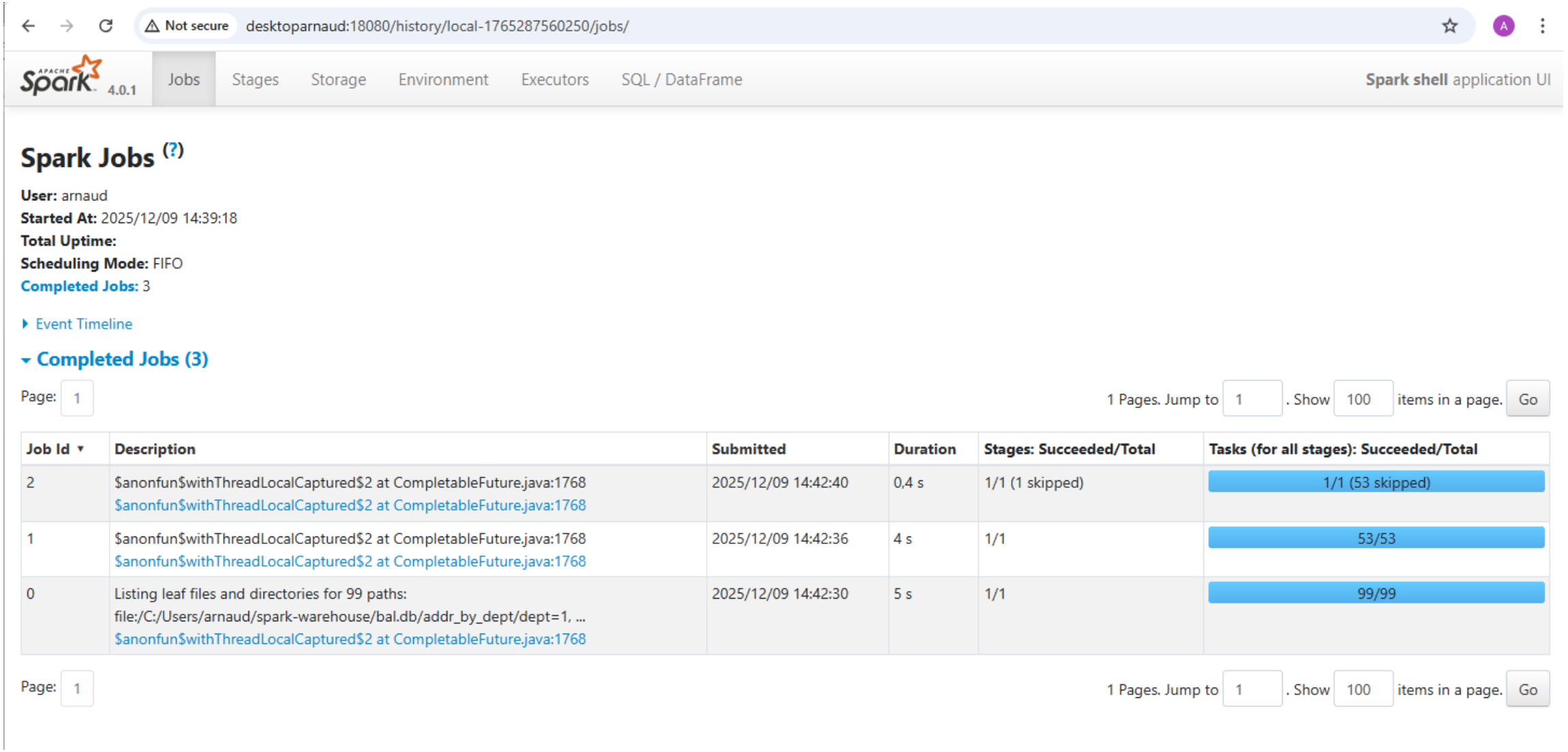
```
25/12/09 15:18:21 INFO JettyUtils: Start Jetty 0.0.0.0:18080 for HistoryServerUI
25/12/09 15:18:21 INFO Utils: Successfully started service 'HistoryServerUI' on port 18080.
25/12/09 15:18:21 INFO HistoryServer: Bound HistoryServer to 0.0.0.0, and started at http://DesktopArnaud:18080
25/12/09 15:18:21 INFO FsHistoryProvider: Parsing file:/C:/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250 for listing data...
25/12/09 15:18:22 INFO FsHistoryProvider: Finished parsing file:/C:/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
```



Incomplete / Complete Applications

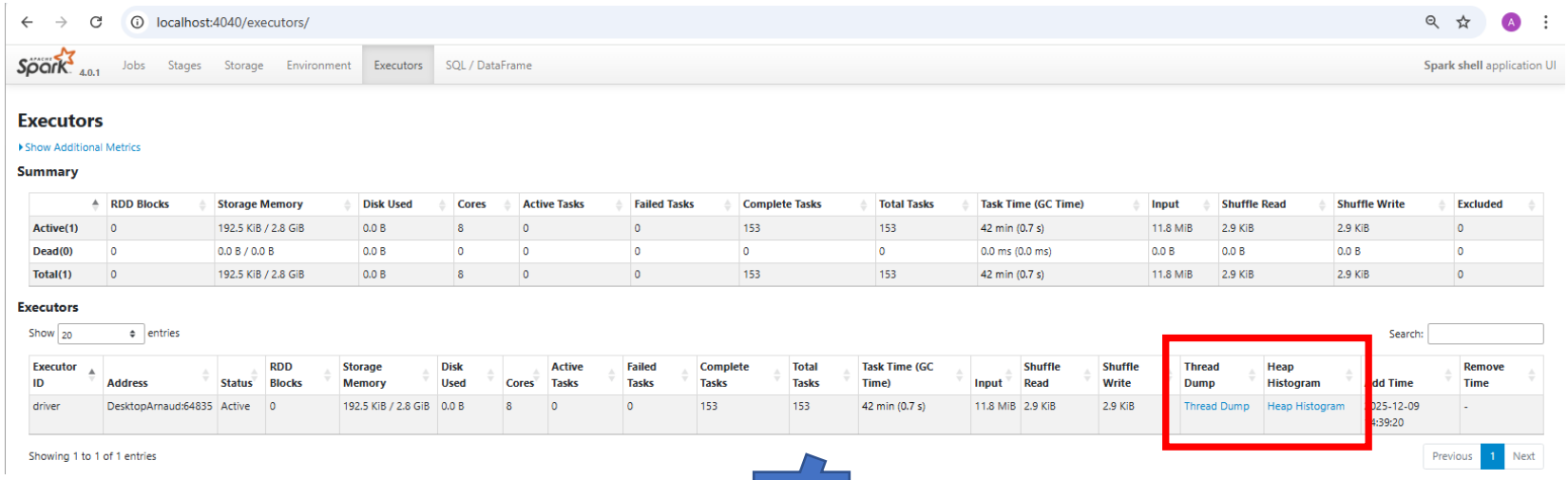


http://localhost:18080/history/${appId}



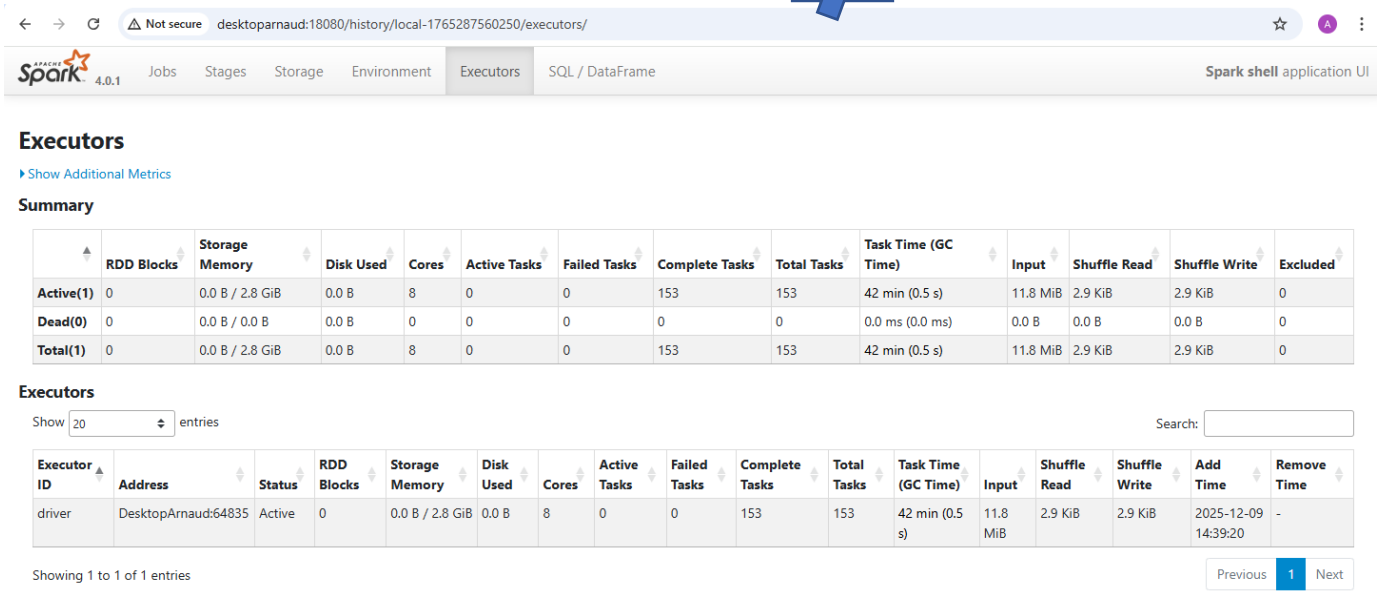
Things You Can NOT Do in Spark History Server … ThreadDumps, Kill Task

Spark UI (live app)



≠

Spark HistoryServer (Offline)



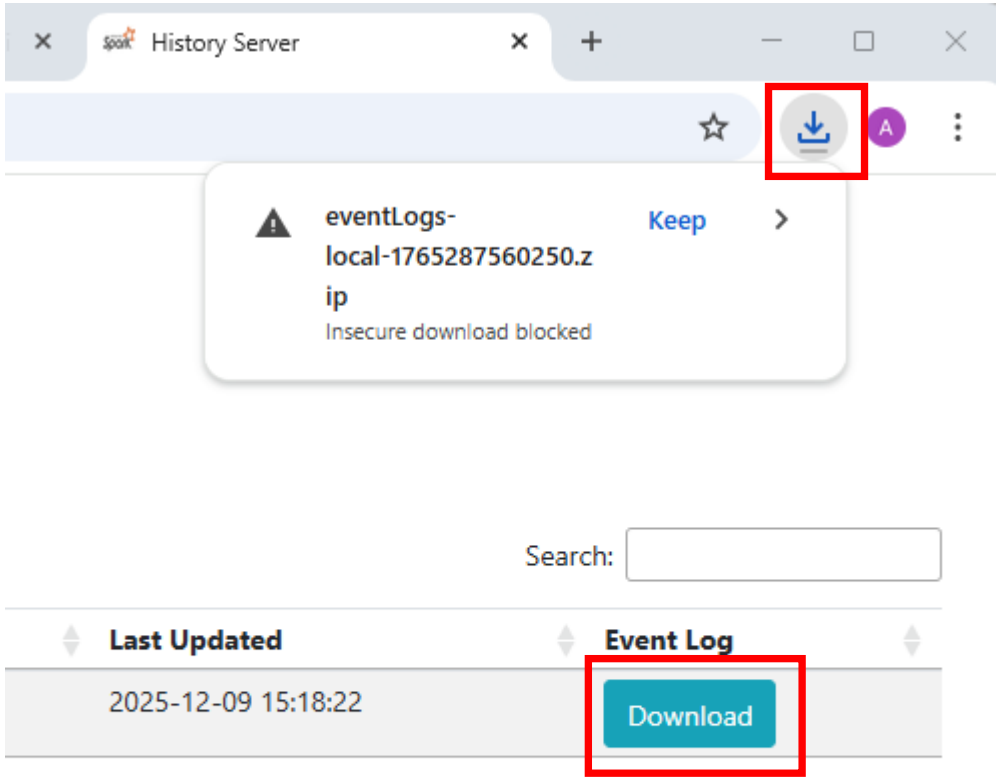
Tip&Tricks for EventLog : Download them

On PROD clusters, eventLogs Files are removed after ~3days
(and may be rewritten, by purging low-level « Task » events)

Download them !!

⇒ for later analyses

⇒ for Requesting Assistance (no credentials / access required)



Spark History Server & Spark UI Limitations

SparkHistory can do OutOfMemoryError while reloading huge spark eventLog files !!

Both SparkUI & Spark History produce amazingly large & complex web pages
With tones of arrows and blue boxes…

Keep only the last 10 000 spark events, and ignore all others !!

Impossible to Analyse « offline » a very long-running application
(need to manually uncompress + prune spark events …)

Questions ?

… Next Steps ?